



Pyrimidine-conjugated fluoroquinolones as new potential broad-spectrum antibacterial agents

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ABSTRACT

Pyrimidine-conjugated fluoroquinolones were constructed to cope with the dreadful resistance. Most of the target pyrimidine derivatives effectively suppressed the growth of the tested strains, especially, 4-aminopyrimidinyl compound **1c** showed a broad antibacterial spectrum and low cytotoxicity and exhibited superior antibacterial potency against *Enterococcus faecalis* with a low MIC of 0.25 µg/mL to norfloxacin and ciprofloxacin. The active compound **1c** with fast bactericidal potency could inhibit the formation of biofilms and showed much lower trend for the development of drug-resistance than norfloxacin and ciprofloxacin. Further exploration revealed that compound **1c** could prompt ROS accumulations in bacterial cells and interact with DNA to form a DNA-**1c** complex, thus facilitating bacterial death. ADME analysis indicated that compound **1c** possessed favorable drug-likeness and promising pharmacokinetic properties. These results demonstrated that pyrimidine-conjugated fluoroquinolones held hope as potential antibacterial candidates and deserve further study.

Quinolones are one of the most important first-line antibacterial agents with creditable biological activity and strong tissue permeability.¹ Many quinolones are continuously developed and successfully used in clinic since their inception, making great contributions to the protection of human health.² However, due to the over-extensive clinical application of quinolones, the bacterial resistance to these antibiotics is increasing, leading to a decline in clinical efficacy.³ Structural modifications on the skeleton of quinolones have become one of the important approaches to combat this urgency. Much work about the modification of N-1, C-3 or C-7 position of classical quinolone backbone was carried out.⁴ Especially the C-7 position of quinolone has been extensively modified by some unique functional groups due to large effect on antibacterial spectrum.⁵ Studies revealed that the increasing bulkiness at C-7 position seemed to be conducive to slacking the drug-resistant development tendency of wild-type bacteria and enhancing the inhibitory potency against some drug-resistant strains.⁶ Fluorine atom in the quinolone skeleton can further increase the tissue permeability of quinolone drugs and piperazine is identified as one of the privileged structures of C-7 position in fluoroquinolones such as

norfloxacin and ciprofloxacin, so piperazine is always utilized as a flexible linker to explore more novel quinolone derivatives with other functional fragments to improve activity and resist the current severe drug-resistance.⁷ In our previous work, several heterocycles such as imidazole,⁸ triazole,⁹ thiazole¹⁰ and benzimidazole¹¹ were introduced to the piperazine terminal to produce promising new fluoroquinolone molecules with excellent antibacterial potential. Pyrimidine as a unique structural heterocycle is prevalently present in numerous clinical drugs¹² especially in antibacterial sulfonamides such as sulfadiazine, sulfamerazine and so on, in which pyrimidine plays important roles in exerting antibacterial activities. However, the combination of pyrimidines with quinolones is rarely reported,¹³ thus the research on introducing pyrimidine into the quinolone skeleton to find more effective antibacterial molecules with lower tendency to trigger drug resistance needs to be vigorously promoted.

Given the above situation and as an extension of our studies on the C-7 position of fluoroquinolones, it is very interested in combining different substituted pyrimidines with fluoroquinolones to produce a new series of pyrimidine-conjugated fluoroquinolones as potential

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antimicrobial agents with multi-targeting possibilities, thus reducing bacterial resistance (Fig. 1). The antibacterial activity of the synthesized pyrimidine derivatives was evaluated, and the possibility of the highly active molecule as new antimicrobial agent was preliminarily investigated through the evaluation of drug resistance development, and ADME research. Moreover, the possible new antibacterial mechanism was explored by oxidative stress and DNA interaction experiments.

Structures of target pyrimidine-conjugated fluoroquinolones **1a–q** were shown in Fig. 2. The reaction of 2-chloropyrimidines with commercial quinolones was carried out by a facile one-pot process with yields of 36.3–75.7 %. It was found that the reaction rates of pyrimidine compounds with electron-withdrawing groups were faster than that of electron-donating substituents, which might be assigned to the former leading to stronger positive electricity at the C-2 position of pyrimidine and lower activation energy of C-Cl bond breaking. The detailed syntheses, relevant spectra and physical characteristic data were available in the Supplementary data.

The antimicrobial efficacies of pyrimidine-conjugated fluoroquinolones **1a–q** were tested by serial dilution method and the results were listed as the minimum inhibitory concentration values (MICs, $\mu\text{g/mL}$) in Table 1 and Table S1. The majority of target new conjugates could effectively suppress the growth of the tested strains, and some prepared compounds were even superior to the reference clinical drugs norfloxacin and ciprofloxacin. Towards methicillin-resistant *Staphylococcus aureus* (MRSA), pyrimidine-conjugated norfloxacin **1a**, **1c**, **1f** and **1g** gave comparable or better antibacterial activity than norfloxacin (MIC = 8 $\mu\text{g/mL}$) and ciprofloxacin (MIC = 2 $\mu\text{g/mL}$). Most pyrimidine-norfloxacin conjugates exerted significant inhibitory effects on *E. faecalis*, especially 4-aminopyrimidinyl derivative **1c** which showed a potent antibacterial potency with an extremely low MIC of 0.25 $\mu\text{g/mL}$, being 16-fold to norfloxacin and 4-fold to ciprofloxacin. The strains of *S. aureus* ATCC 25,923 and *S. aureus* ATCC 29,213 were only sensitive to compound **1c**, and the inhibitory abilities were 4 and 8 times higher than norfloxacin, respectively. Except for pyrimidine-norfloxacin conjugate **1a** and 4-trifluoromethylpyrimidine-norfloxacin compound **1d**, other pyrimidine-norfloxacin conjugates showed moderate to good suppression on *S. aureus* with MIC values between 1 and 8 $\mu\text{g/mL}$. Moreover, it was found that most pyrimidine-norfloxacin with electron-donating groups could effectively inhibit the growth of *K. pneumonia*, but all of the pyrimidine-norfloxacin had unsatisfactory antibacterial activity against *E. coli*. Notably, 4-aminopyrimidinyl derivative **1c** also gave antibacterial potency towards Gram-negative strains *P. aeruginosa* and *A. baumannii*.

Another commercial fluoroquinolone ingredient ciprofloxacin was chosen to conjugate with pyrimidines, which produced pyrimidine-ciprofloxacin hybrids **1j–q**. The pyrimidinyl derivative **1j** without

substituent in pyrimidine ring showed moderate to good antibacterial activities against Gram-positive bacteria, and excellent anti-*E. coli* ATCC 25,922 potency with a MIC value of 1 $\mu\text{g/mL}$. Trifluoromethyl group, which is widely present in clinical drugs and is extensively employed to design new drug molecules, was introduced into pyrimidinyl conjugate **1l** with the hope to produce molecules with excellent antibacterial activities, but the result was to no avail. Methoxypyrimidine derivative **1o** was found to suppress the growth of *A. baumannii*, with the same MIC of 1 $\mu\text{g/mL}$ as ciprofloxacin. Although 5-fluoropyrimidine-derived norfloxacin conjugate **1h** could not inhibit the growth of the tested strains, its ciprofloxacin conjugate **1p** exhibited desirable anti-MRSA activity with a MIC value of 1 $\mu\text{g/mL}$, which was 8- and 2-fold in comparison to norfloxacin and ciprofloxacin, respectively.

A lot of pyrimidine compounds such as voriconazole also exhibit antifungal potential, so *in vitro* antifungal results were evaluated (Table S1). All pyrimidine-norfloxacin conjugates **1a–i** gave worse inhibitory activities against *C. albicans* and *C. parapsilosis* ATCC 22019 than fluconazole. As for *A. fumigatus*, pyrimidine derivatives **1a**, **1c**, **1e** and **1g** showed obvious inhibitory effects with MIC values of 2–4 $\mu\text{g/mL}$. Pyrimidine-conjugated ciprofloxacin **1j** exhibited excellent anti-*A. fumigatus* activity with the MIC value of 1 $\mu\text{g/mL}$, being more potent than the reference drug fluconazole (MIC = 256 $\mu\text{g/mL}$). Both 5-ethylpyrimidine and 5-methoxypyrimidine conjugates **1n** and **1o** were particularly prominent against *C. tropicalis* with the same MIC value of 2 $\mu\text{g/mL}$. In addition, methoxypyrimidine-ciprofloxacin **1o** also displayed superior effect on *C. parapsilosis* ATCC 22019 with MIC value of 2 $\mu\text{g/mL}$ to fluconazole (MIC = 4 $\mu\text{g/mL}$), indicating that the introduction of methoxyl group was conducive to inhibiting fungal growth.

The cytotoxicity was necessary to be determined as a crucial aspect of druggability explorations.¹⁴ Hence, cell viability with the treatment of the highly active compounds **1c** and **1j** was tested by the CCK-8 assay. As shown in Fig. 3, the viable cells exceed 90% at 12.5 $\mu\text{g/mL}$ ($50 \times \text{MIC}$) after the treatment of 4-aminopyrimidine-conjugated quinolone **1c**, and as even the concentration reached up to 50 $\mu\text{g/mL}$ ($200 \times \text{MIC}$), about 70% HaCat cells remained viable. The similar result was also observed on the compound **1j** with potent antifungal activity. The viability rate of HaCat cells kept about 70% at 50 $\mu\text{g/mL}$ ($50 \times \text{MIC}$). Above results indicated that these pyrimidine-conjugated fluoroquinolones possessed good biosafety and their antimicrobial activities were not due to the general toxicity.

According to the above data, the introduction of pyrimidines could not only enhance the antibacterial activities of fluoroquinolones towards some tested strains but also lead antifungal potential. Notably, 4-aminopyrimidine-conjugated quinolone **1c** displayed a broad antibacterial spectrum (MICs = 0.25–16 $\mu\text{g/mL}$) and exerted potent inhibition efficiency to *E. faecalis* with a low MIC of 0.25 $\mu\text{g/mL}$ and low cytotoxicity, which was worth being selected as a representative to further do the preliminary evaluation of druggability and explore the antibacterial mechanism.

The persistence and refractory of bacterial infection caused by drug resistance pose an indelible impact on global health,¹⁵ and hence, the resistance-induced tendency of candidates is an important evaluation indicator for being developed and listed.¹⁶ The propensity of drug resistance induced by aminopyrimidine-conjugated quinolone **1c** towards *E. faecalis* was evaluated using norfloxacin and ciprofloxacin as positive controls, and the development of drug resistance was displayed in Fig. 4. The MICs of compound **1c** against *E. faecalis* remained stable within 11 days and merely occurred a very low enhancement at the end of evaluation, which manifested that compound **1c** possessed a low tendency to trigger the development of resistance. However, the fold-change in the MICs of both clinical norfloxacin and ciprofloxacin showed 7-fold increases, suggesting that the bacterial resistances caused by norfloxacin and ciprofloxacin were significant. The low trend in the development of *E. faecalis* resistance to aminopyrimidine-conjugated quinolone **1c** was another advantage in addition to its strong antibacterial activity, being compared to available antibacterial drugs.

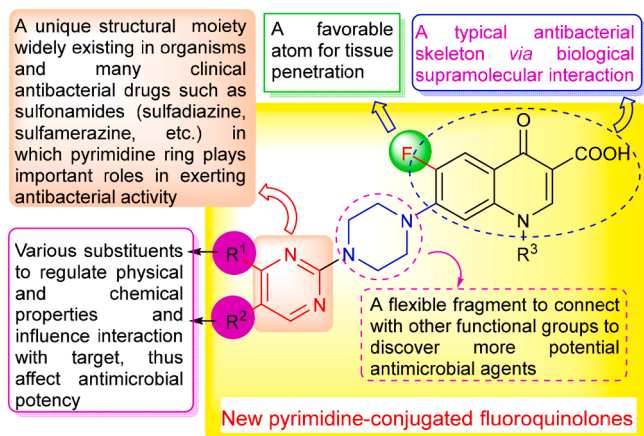


Fig. 1. Design of pyrimidine-conjugated fluoroquinolones as antibacterial agents.

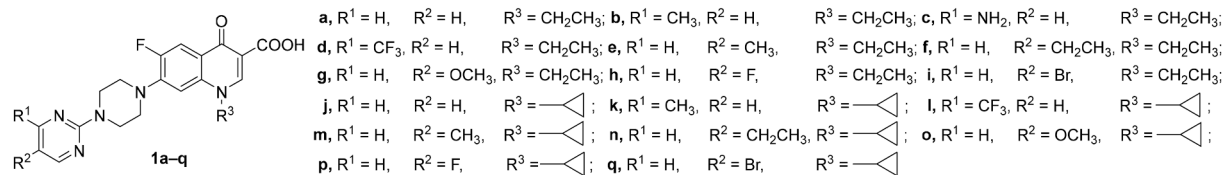
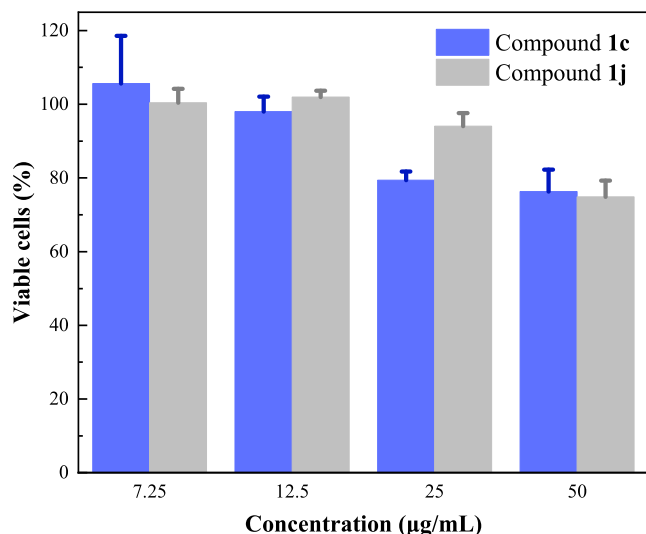
Fig. 2. Structures of target pyrimidine-conjugated fluoroquinolones **1a–q**.

Table 1

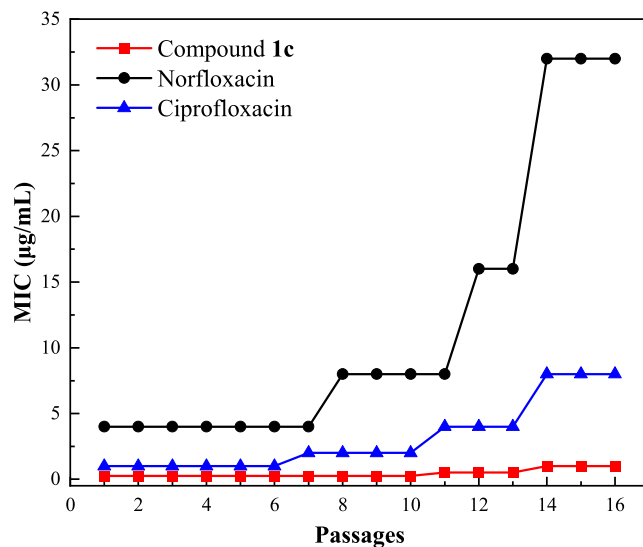
Antibacterial data as MICs (μg/mL) for pyrimidine-conjugated fluoroquinolones **1a–q** and reference drugs.^a

Compounds	Gram-positive bacteria					Gram-negative bacteria					
	MRSA	<i>E. F.</i>	<i>S. A.</i>	<i>S. A.</i> 25923	<i>S. A.</i> 29213	<i>K. P.</i>	<i>E. C.</i>	<i>E. C.</i> 25922	<i>P. A.</i>	<i>P. A.</i> 27853	<i>A. B.</i>
1a	2	8	64	8	32	4	16	64	16	128	64
1b	32	1	4	4	16	4	64	1	128	16	256
1c	1	0.25	1	0.5	0.5	0.5	4	16	0.25	0.5	4
1d	> 256	64	> 256	256	128	128	> 256	256	128	> 256	128
1e	8	2	4	64	4	2	16	2	4	128	128
1f	2	2	8	2	64	2	16	1	2	128	128
1g	1	1	2	32	8	64	64	2	32	128	128
1h	64	4	4	16	16	128	16	128	32	256	> 256
1i	128	64	8	64	64	256	64	128	8	64	64
1j	16	2	4	64	32	64	8	1	8	32	> 256
1k	64	64	256	4	32	64	64	8	64	64	2
1l	64	8	128	128	64	128	128	32	64	128	> 256
1m	256	4	128	2	64	> 256	64	8	2	256	128
1n	128	64	1	8	64	256	8	4	4	64	64
1o	32	32	256	8	2	16	64	4	64	32	1
1p	1	8	8	8	32	128	128	8	64	256	> 256
1q	128	128	16	16	8	128	64	128	16	32	32
Norfloxacin	8	4	4	2	4	4	8	8	4	4	2
Ciprofloxacin	2	1	2	2	0.5	2	1	2	0.5	1	1

^a MRSA, methicillin-resistant *Staphylococcus aureus* N315; *E. F.*, *Enterococcus faecalis* S. A., *Staphylococcus aureus*; *S. A.* 25923, *Staphylococcus aureus* ATCC 25923; *S. A.* 29213, *Staphylococcus aureus* ATCC 29213; *K. P.*, *Klebsiella pneumoniae*; *E. C.*, *Escherichia coli*; *E. C.* 25922, *Escherichia coli* ATCC 25922; *P. A.*, *Pseudomonas aeruginosa*; *P. A.*, 27853, *Pseudomonas aeruginosa* ATCC 27853; *A. B.*, *Acinetobacter baumannii*.

Fig. 3. Cytotoxicity of highly active compounds **1c** and **1j** on normal HaCat cells.

The rapid bactericidal capacity of antibacterial agents is conducive to shortening the treatment time and reducing the possibility of drug resistance,¹⁷ so the bactericidal ability of compound **1c** (1 and 4 × MIC) was conducted with norfloxacin and ciprofloxacin as positive controls. Aminopyrimidinyl derivative **1c** at 1 × MIC value caused a 2.5 log reduction in bacterial colony count of *E. faecalis* within 6 h (Fig. 5), which was comparable to norfloxacin and slightly weaker than ciprofloxacin. Additionally, 4 × MIC compound **1c** achieved a more than 3.5

Fig. 4. Development of *E. faecalis* resistance to aminopyrimidine-conjugated quinolone **1c**, norfloxacin and ciprofloxacin.

log reduction within 6 h at the same concentration as ciprofloxacin (1 μg/mL). These results revealed that compound **1c** possessed an accelerating sterilization rate with the increase of content, which might contribute to the low resistance of *E. faecalis* towards compound **1c**.

Biofilm obstructs the diffusion and penetration of antibacterial agents, resulting in the reduced efficiency or even failure of existing drugs, and then stimulates the aggravation of drug resistance.¹⁸

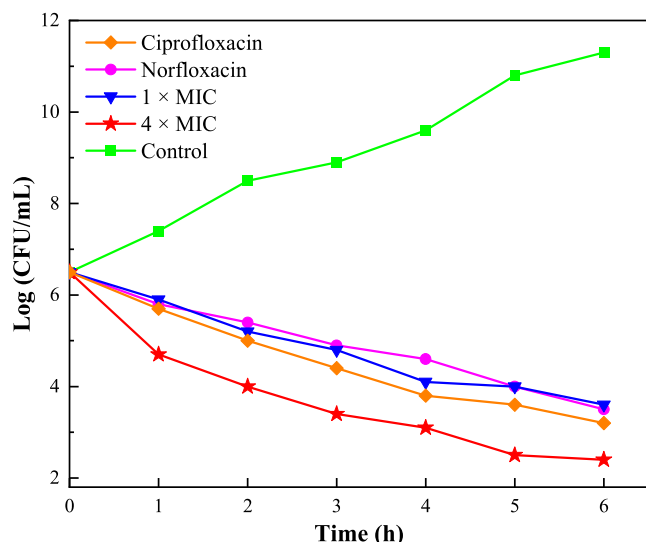


Fig. 5. Bactericidal kinetic of aminopyrimidine-conjugated quinolone 1c (1 and 4 × MIC), norfloxacin (1 × MIC) and ciprofloxacin (1 × MIC) towards *E. faecalis*.

Therefore, the efficacy of aminopyrimidinyl derivative 1c on biofilm disturbance was evaluated with crystal violet.¹⁹ As depicted in Fig. 6, compound 1c could effectively impede the biofilm formation at the concentration of 2 µg/mL (8 × MIC) with an inhibition rate of 53.9%, and it presented a dose-dependent increasing inhibited effect, which could feed a feasible execution for the low drug-resistance trend.

Oxidative stress caused by high levels of intracellular ROS may induce the destruction of cytoskeleton and autophagy, including the destruction of membrane and DNA.²⁰ It has been reported that aromatic pyrimidines might be conducive to the production of ROS, which could hinder the survival of microorganisms by stimulating the accumulation of ROS.²¹ Therefore, the ROS accumulation in *E. faecalis* treated with compound 1c was evaluated with fluorescence dye dichlorofluorescein diacetate (DCFH-DA).²² Nonfluorescent DCFH-DA could be oxidized to fluorescent dichlorofluorescein (DCF) in the presence of ROS and the fluorescence intensity could act as an oxidation indicator.²³ According to Fig. 7, compound 1c could potentiate the accumulation of ROS in bacterial cells and there was about 2.5-fold amplification at 8 × MIC in fluorescence intensity with respect to the control group. These results

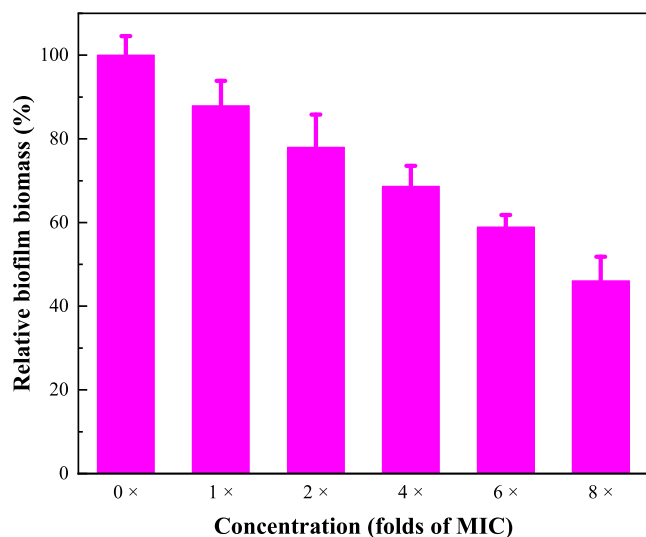


Fig. 6. *E. faecalis* biofilm inhibition by aminopyrimidine-conjugated quinolone 1c.

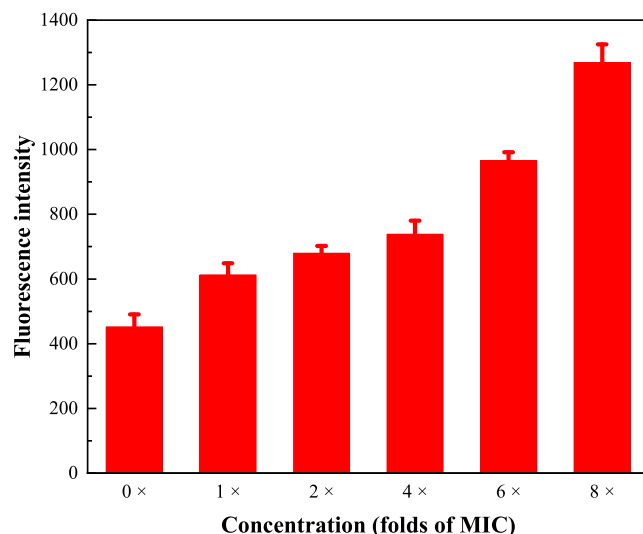


Fig. 7. Intracellular ROS accumulation in *E. faecalis* treated with aminopyrimidine-conjugated quinolone 1c.

suggested that compound 1c could promote the accumulation of intracellular ROS and cause oxidative imbalance, contributing to raising antibacterial activity.

DNA undertakes the function of storing and transmitting genetic information²⁴ and is an important target in the design and development of new drugs²⁵. Pyrimidines with two nitrogen atoms were described to facilitate interaction with DNA in literature. Hence, the interaction model between aminopyrimidine-conjugated quinolone 1c and calf thymus DNA (ct-DNA) was explored by distinguishing the hypochromism and hyperchromism of DNA UV-Vis spectrum.²⁶ As displayed in Fig. 8A, the maximum absorption peak at 260 nm of DNA strengthened and an apparent red shift was observed with the increasing contents of compound 1c. Furthermore, the absorption sum value of free DNA and free molecule 1c was weaker than that of the binary 1c-DNA complex, therefore a weak hyperchromism came to light (Fig. 8B). The reason might be the formation of DNA-1c complex, contributing to the partial unwinding of DNA double helix structure.²⁷ The binding constant ($K = 1.01 \times 10^5$ L/mol) also indicated that aminopyrimidinyl derivative 1c showed a strong DNA-binding ability (Fig. S1).

Further study on the interactive approach between aminopyrimidine-conjugated quinolone 1c and DNA was carried out by competitive experiments with acridine orange (AO),²⁸ and the result was shown in Fig. 9. An obvious reduction was observed in the fluorescence intensity of AO-DNA complex around 530 nm with the concentration of hybrid 1c increasing, indicating that aminopyrimidinyl derivative 1c could inset into DNA by replacing AO in the AO-DNA complex. Moreover, an intuitively docking simulation was exhibited in Fig. S2, which provided evidence for the intercalation of compound 1c into DNA. All these exposed the change in DNA duplexes and the formation of DNA – 1c supramolecular complex, thus aminopyrimidine-conjugated quinolone 1c exerting more potent antibacterial potency.

Bacterial DNA gyrase as an exact target of quinolones plays an important role in the normal physiological function of DNA.²⁹ Given the interaction between aminopyrimidine-conjugated quinolone 1c and DNA, a docking simulation between compound 1c and DNA gyrase (PDB code: 4HZ5) was implemented to further determine the interaction site between molecule 1c and DNA. Fig. 10A showed that three oxygen atoms of 3-carboxyl and 4-carbonyl moieties could bind with residue LYS-202 through hydrogen bonds with the distance of 2.2, 2.5 and 2.4 Å. Meanwhile, it was observed that there was a hydrogen bond between the nitrogen atom at the 3-position of pyrimidine and residue GLY-203. The hydrogen atom of NH₂ group possessed a hydrogen bond with residue GLY-203 at a distance of 2.3 Å, which could be responsible for the

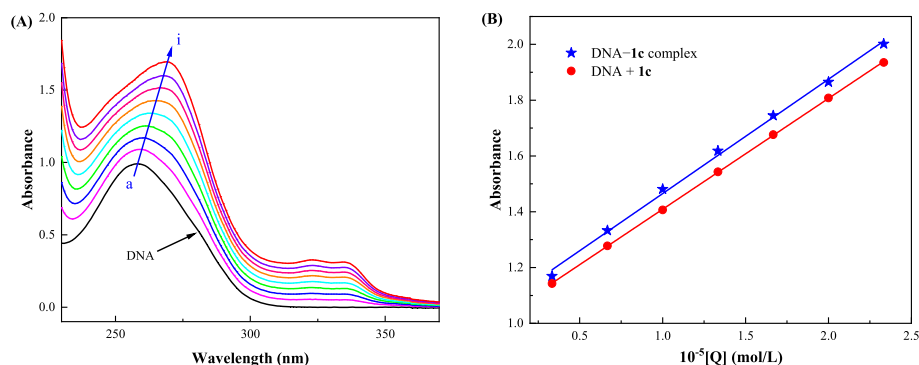


Fig. 8. Interaction of aminopyrimidine-conjugated quinolone **1c** with ct-DNA. (A) Ultraviolet spectra of DNA with incremental compound **1c** (B) Absorption comparison at 260 nm between the **1c** – DNA complex and the sum of DNA and compound **1c**.

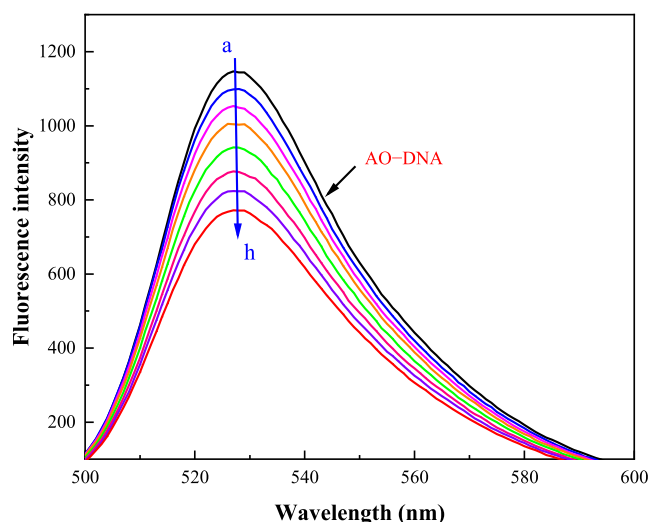


Fig. 9. Competitive effect between aminopyrimidine-conjugated quinolone **1c** and the AO-DNA system.

compound **1c** gave the most potent antibacterial activity among these pyrimidine-conjugated fluoroquinolones. Moreover, the fluoroquinolone skeleton was found to interact with residue LYS-202 through π -cation bond (Fig. 10B). These results suggested that compound **1c** was a multi-targeted antibacterial agent, which might be the reason for the low tendency of drug resistance.

In general, clinical candidates need not only excellent bioactivities but also good pharmacokinetics and bioavailabilities. The pharmacokinetic physicochemical parameters of compound **1c**, norfloxacin and ciprofloxacin were predicted through the online website SwissADME.³⁰ The results (Table 2) showed that compound **1c** could completely accord with Lipinski's rules. Meanwhile, molecule **1c** had a similar bioavailability score and GI (gastrointestinal tract) absorption to norfloxacin and ciprofloxacin and could not penetrate blood-brain barrier (BBB). These results demonstrated that aminopyrimidinyl quinolone **1c** possessed excellent pharmacokinetic profiles and drug-likeness properties.

This work developed a class of pyrimidine-conjugated fluoroquinolones to withstand drug-resistance antimicrobial infections. Bioactivity evaluation revealed that most of the prepared pyrimidine conjugates could effectively suppress the growth of the tested strains. Prominently, 4-aminopyrimidine-conjugated quinolone **1c** showed a

Table 2
ADME evaluations of aminopyrimidine-conjugated quinolone **1c**, norfloxacin and ciprofloxacin.

Compounds	1c	Norfloxacin	Ciprofloxacin
MW (g/mol) < 500	412.42	319.33	331.34
MLog P $\leq$ 4.15	0.73	1.04	1.28
H-bond acceptors < 10	6	5	5
H-bond donors < 5	2	2	2
Rotatable bonds < 10	4	3	3
Lipinski violation	0	0	0
Bioavailability score	0.56	0.55	0.55
GI absorption	High	High	High
BBB permeant	No	No	No

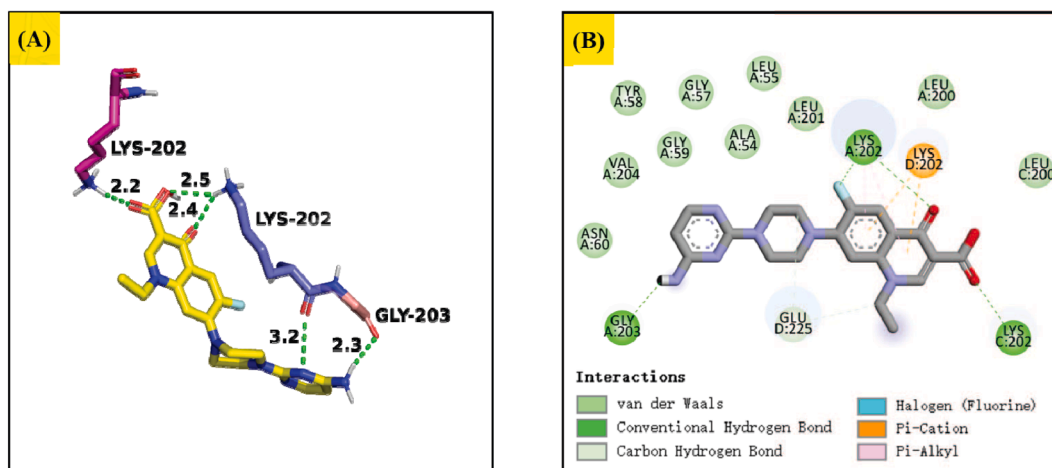


Fig. 10. 3D conformation (A) and 2D conformation (B) of aminopyrimidine-conjugated quinolone **1c** docked in DNA gyrase (PDB code: 4HZ5).

broad spectrum and good biosafety and exerted remarkable antibacterial activity against *E. faecalis* with a MIC value of 0.25 µg/mL. Further studies manifested that compound **1c** possessed rapid bactericidal ability, gave no apparent tendency to trigger drug resistance of *E. faecalis* and could significantly hinder the formation of biofilm. Moreover, derivative **1c** could not only potentiate ROS accumulation to perturb intracellular oxidation-antioxidant balance but also embed into DNA to affect the replication of DNA, thus exerting forceful antibacterial activity. *In silico* ADME study indicated that compound **1c** possessed good pharmacokinetic properties and drug-likeness. These encouraging results manifested that 4-aminopyrimidine-conjugated fluoroquinolone **1c** was promising for further contributing to the development of broad-spectrum antibacterial agents.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bmcl.2022.128885>.

References

- Kumar A, Saxena N, Mehrotra A, et al. Review: Studies on the synthesis of quinolone derivatives with their antibacterial activity (Part 1). *Curr. Org. Chem.* 2020;24: 817–854.
- (a) Pham TDM, Ziora ZM, Blaskovich MAT. Quinolone antibiotics. *Med. Chem. Commun.* 2019;10:1719–1739 (b) Mitscher LA. Bacterial topoisomerase inhibitors: Quinolone and pyridone antibacterial agents. *Chem. Rev.* 2005;105:559–592.
- (a) Chu XM, Wang C, Liu W, et al. Quinoline and quinolone dimers and their biological activities: An overview. *Eur. J. Med. Chem.* 2019;161:101–117 (b) Zhang TT, Shen WY, Liu ML, et al. Synthesis, antimycobacterial and antibacterial activity of quinolone derivatives containing an 3-alkoxyimino-4-(cyclopropylamino)methylpyrrolidine moiety. *Eur. J. Med. Chem.* 2015;104:73–85 (c) Xie YP, Sangaraiha N, Meng JP, et al. Unique carbazole-oxadiazole derivatives as new potential antibiotics for combating Gram-positive and -negative bacteria. *J. Med. Chem.* 2022;65: 6171–6190.
- (a) Cui SF, Peng LP, Zhang HZ, et al. Novel hybrids of metronidazole and quinolones: Synthesis, bioactive evaluation, cytotoxicity, preliminary antimicrobial mechanism and effect of metal ions on their transportation by human serum albumin. *Eur. J. Med. Chem.* 2014;86:318–334 (b) Cui SF, Addla D, Zhou CH. Novel 3-aminothiazolquinolones: Design, synthesis, bioactive evaluation, SARs, and preliminary antibacterial mechanism. *J. Med. Chem.* 2016;59:4488–4510 (c) Cheng Y, Avula SR, Gao WW, et al. Multi-targeting exploration of new 2-aminothiazolyl quinolones: Synthesis, antimicrobial evaluation, interaction with DNA, combination with topoisomerase IV and penetrability into cells. *Eur. J. Med. Chem.* 2016;124:935–945 (d) Gao F, Wang P, Yang H, et al. Recent developments of quinolone-based derivatives and their activities against *Escherichia coli*. *Eur. J. Med. Chem.* 2018;157: 1223–1248.
- (a) Zhang GF, Zhang S, Pan BF, et al. 4-Quinolone derivatives and their activities against Gram positive pathogens. *Eur. J. Med. Chem.* 2018;143:710–723 (b) Malik M, Mustaez A, Schwanz HA, et al. *Nucleic. Acids. Res.* 2016;44:3304–3316.
- Fedorowicz J, Sączewski J. Modifications of quinolones and fluoroquinolones: Hybrid compounds and dual-action molecules. *Monatsh. Chem.* 2018;149: 1199–1245.
- Gao C, Fan YL, Zhao F, et al. Quinolone derivatives and their activities against methicillin-resistant *Staphylococcus aureus* (MRSA). *Eur. J. Med. Chem.* 2018;157: 1081–1095.
- (a) Zhang L, Kumar KV, Geng RX, et al. Design and biological evaluation of novel quinolone-based metronidazole derivatives as potent Cu²⁺ mediated DNA-targeting antibacterial agents. *Bioorg. Med. Chem. Lett.* 2015;25:3699–3705 (b) Zhang L, Peng XM, Damu GLV, et al. Comprehensive review in current developments of imidazole-based medicinal chemistry. *Med. Res. Rev.* 2014;34:340–437.
- Zhang L, Kumar KV, Rasheed S, et al. Design, synthesis, and antibacterial evaluation of novel azolylthioether quinolones as MRSA DNA intercalators. *Med. Chem. Commun.* 2015;6:1303–1310.
- Chen JP, Battini N, Ansari MF, et al. Membrane active 7-thiazoxime quinolones as novel DNA binding agents to decrease the genes expression and exert potent anti-methicillin-resistant *Staphylococcus aureus* activity. *Eur. J. Med. Chem.* 2021;217, 113340.
- (a) Zhang L, Addla D, Ponmani J, et al. Discovery of membrane active benzimidazole quinolones-based topoisomerase inhibitors as potential DNA-binding antimicrobial agents. *Eur. J. Med. Chem.* 2016;111:160–182 (b) Zhang L, Kumar KV, Rasheed S, et al. Design, synthesis, and antimicrobial evaluation of novel quinolone imidazoles and interactions with MRSA DNA. *Chem. Biol. Drug. Des.* 2015;86:648–655 (c) Wang YN, Bheemanaboina RRY, Gao WW, et al. Discovery of benzimidazole-quinolone hybrids as new cleaving agents toward drug-resistant *Pseudomonas aeruginosa* DNA. *ChemMedChem.* 2018;13:1004–1017.
- (a) Liu HB, Gao WW, Tangdanchu VKR, et al. Novel aminopyrimidinyl benzimidazoles as potentially antimicrobial agents: Design, synthesis and biological evaluation. *Eur. J. Med. Chem.* 2018;143: 66–84 (b) Sui YF, Li D, Wang J, et al. Design and biological evaluation of a novel type of potential multi-targeting antimicrobial sulfanilamide hybrids in combination of pyrimidine and azoles. *Bioorg. Med. Chem. Lett.* 2020;30:126982 (c) Zhou Y, Zhang SS, Cai M, et al. Design, synthesis, and antifungal activity of 2,6-dimethyl-4-aminopyrimidine hydrazones as PDHc-E1 inhibitors with a novel binding mode. *J. Agric. Food Chem.* 2021;69:5804–5817 (d) Sui YF, Ansari MF, Zhou CH. Pyrimidinetrione-imidazoles as a unique structural type of potential agents towards *Candida albicans*: Design, synthesis and biological evaluation. *Chem. Asian. J.* 2021;16:1417–1429.
- Majid ME, Hossein E, Marzieh A, et al. Synthesis and antibacterial evaluation of new pyrimidyl N-ciprofloxacin derivatives. *ChemistrySelect.* 2019;4:8930–8933.
- (a) Yang XC, Hu CF, Zhang PL, et al. Coumarin thiazoles as unique structural skeleton of potential antimicrobial agents. *Bioorg. Chem.* 2022;124:105855 (b) Dewangan RP, Verma DP, Verma NK, et al. Spermine-conjugated short proline-rich lipopeptides as broad-spectrum intracellular targeting antibacterial agents. *J. Med. Chem.* 2022;65: 5433–5448.
- (a) Yilmaz NK, Schiffer CA. Introduction: Drug resistance. *Chem. Rev.* 2021;121: 3235–3237 (b) Varela MF, Stephen J, Lekshmi M, et al. Bacterial resistance to antimicrobial agents. *Antibiotics* 2021;10:593 (c) Laxminarayan R. The overlooked pandemic of antimicrobial resistance. *The Lancet* 2022;399:606–607.
- (a) Li HX, Liu JY, Liu CF, et al. Design, synthesis, and biological evaluation of membrane-active bakuchiol derivatives as effective broad-spectrum antibacterial agents. *J. Med. Chem.* 2021;64:5603–5619 (b) Li FF, Zhao WH, Tangdanchu VKR, et al. Discovery of novel phenylhydrazine-based oxindole-thiazoles as potent antibacterial agents toward *Pseudomonas aeruginosa*. *Eur. J. Med. Chem.* 2022;239: 114521.
- (a) Sui YF, Ansari MF, Fang B, et al. Discovery of novel purinylthiazolylethanone derivatives as anti-*Candida albicans*. agents through possible multi-faceted mechanisms. *Eur. J. Med. Chem.* 2021;221:113557 (b) Sun H, Huang SY, Jeyakkumar P, et al. Natural berberine-derived azolyl ethanol as new structural antibacterial agents against drug-resistant *Escherichia coli*. *J. Med. Chem.* 2022;65:436–459 (c) Liu JY, Li HX, Li HZ, et al. Rational design of dipicolylamine-containing carbazole amphiphiles combined with Zn²⁺ as potent broad-spectrum antibacterial agents with a membrane-disruptive mechanism. *J. Med. Chem.* 2021;64:10429–10444 (d) Deng Z, Sun H, Bheemanaboina RRY, et al. Natural aloe emodin-hybridized sulfonamide aminophosphates as novel potential membrane-perturbing and DNA-intercalating agents against *Enterococcus faecalis*. *Bioorg. Med. Chem. Lett.* 2022;64:128695.
- (a) Van-Wolferen M, Orell A, Albers SV. Archaeal biofilm formation. *Nat. Rev. Microbiol.* 2018;16:699–713 (b) Mukherjee M, Cao B. Engineering controllable biofilms for biotechnological applications. *Microb. Biotechnol.* 2020;14:74–78 (c) Vishwakarma A, Dang F, Ferrell A, et al. Peptidomimetic polyurethanes inhibit bacterial biofilm formation and disrupt surface established biofilms. *J. Am. Chem. Soc.* 2021;143:9440–9449.
- (a) Wang J, Ansari MF, Zhou CH. Identification of unique quinazoline thiazoles as novel structural scaffolds for potential Gram-negative bacterial conquerors. *J. Med. Chem.* 2021;64:7630–7645 (b) Wang J, Zhang PL, Ansari MF, et al. Molecular design and preparation of 2-aminothiazole sulfanilamide oximes as membrane active antibacterial agents for drug resistant *Acinetobacter baumannii*. *Bioorg. Chem.* 2021; 113:105039 (c) Xie YP, Ansari MF, Zhang SL, et al. Novel carbazole-oxadiazoles as potential *Staphylococcus aureus* germicides. *Pestic Biochem Phys.* 2021;175:104849 (d) Lencova S, Svarcova V, Stiborova H, et al. Bacterial biofilms on polyamide nanofibers: Factors influencing biofilm formation and evaluation. *ACS Appl. Mater. Interfaces.* 2021;13:2277–2288.
- (a) Zhang HY, Lu QR, Zhang J, et al. Discovery of novel nitrogenous heterocyclic-containing quinoxaline,1,4-di-N-oxides as potent activator of autophagy in *M.tb*-infected macrophages. *Eur. J. Med. Chem.* 2021;223:113657 (b) Mohammad YW, Aijaz A, Faisal MA, et al. Citral derivative activates cell cycle arrest and apoptosis signaling

- pathways in *Candida albicans* by generating oxidative stress. *Bioorg. Chem.* 2021;115: 105260.
- 21 (a) Yang X, Sun H, Maddili SK, et al. Dihydropyrimidinone imidazoles as unique structural antibacterial agents for drug-resistant gram-negative pathogens. *Eur. J. Med. Chem.* 2022;232:114188 (b) Suresh L, Kumar PSV, Poornachandra Y, et al. Design, synthesis and evaluation of novel pyrazolo-pyrimido[4,5-d]pyrimidine derivatives as potent antibacterial and biofilm inhibitors. *Bioorg. Med. Chem. Lett.* 2017;27:1451–1457.
 - 22 (a) Donthiboina K, Anchi P, Gurram S, et al. Synthesis and biological evaluation of substituted *N*-(2-(1*H*-benzo[d]imidazol-2-yl)phenyl)cinnamides as tubulin polymerization inhibitors. *Bioorg. Chem.* 2020;103:104191 (b) Yang XC, Zhang PL, Kumar KV, et al. Discovery of unique thiazolidinone-conjugated coumarins as novel broad spectrum antibacterial agents. *Eur. J. Med. Chem.* 2022;232:114192 (c) Zhang PL, Lavanya G, Zhang SL, et al. An unanticipated discovery towards novel naphthalimide corbelled aminothiazoximes as potential anti-MRSA agents and allosteric modulators for PBP2a. *Eur. J. Med. Chem.* 2021;229:114050 (d) Sun H, Ansari MF, Fang B, et al. Natural berberine-hybridized benzimidazoles as novel unique bactericides against *Staphylococcus aureus*. *J. Agric. Food Chem.* 2021;69: 7831–7840.
 - 23 Suresh L, Kumar PSV, Poornachandra Y, et al. An efficient one-pot synthesis of thiochromenof[3,4-*d*]pyrimidines derivatives: Inducing ROS dependent antibacterial and anti-biofilm activities. *Bioorg. Chem.* 2016;68:159–165.
 - 24 (a) Chen XY, Gonçalves MAFV. DNA, RNA, and protein tools for editing the genetic information in human cells. *iScience* 2018;6:247–263 (b) Dong YH, Yao C, Zhu Y, et al. DNA functional materials assembled from branched DNA: Design, synthesis, and applications. *Chem. Rev.* 2020;120:9420–9481.
 - 25 (a) Hu CF, Zhang PL, Sui YF, et al. Ethylenic conjugated coumarin thiazolidinediones as new efficient antimicrobial modulators against clinical methicillin-resistant *Staphylococcus aureus*. *Bioorg. Chem.* 2020;94:103434 (b) Xie JY, Zhou M, Qian YX, et al. Addressing MRSA infection and antibacterial resistance with peptoid polymers. *Nat. Commun.* 2021;12:5898 (c) Wang J, Ansari MF, Lin JM, et al. Design and synthesis of sulfanilamide aminophosphonates as novel antibacterial agents towards *Escherichia coli*. *Chin. J. Chem.* 2021;39:2251–2263.
 - 26 (a) Wang J, Ansari MF, Zhou CH. Unique para-aminobenzenesulfonyl oxadiazoles as novel structural potential membrane active antibacterial agents towards drug-resistant methicillin resistant *Staphylococcus aureus*. *Bioorg. Med. Chem. Lett.* 2021;41: 127995 (b) Kahrman N, Peker K, Serdaroglu V, et al. Novel 2-amino-4-aryl-6-pyridopyrimidines and *N*-alkyl derivatives: Synthesis, characterization and investigation of anticancer, antibacterial activities and DNA/BSA binding affinities. *Bioorg. Chem.* 2020;99:103805 (c) Wang LL, Battini N, Bheemanaboina RRY, et al. Design and synthesis of aminothiazolyl norfloxacin analogues as potential antimicrobial agents and their biological evaluation. *Eur. J. Med. Chem.* 2019;167:105–123 (d) Wang LL, Battini N, Bheemanaboina RRY, et al. A new exploration towards aminothiazolquinolone oximes as potentially multi-targeting antibacterial agents: Design, synthesis and evaluation acting on microbes, DNA, HSA and topoisomerase IV. *Eur. J. Med. Chem.* 2019;179:166–181.
 - 27 (a) Wang J, Battini N, Ansari MF, et al. Synthesis and biological evaluation of quinazolonethiazoles as new potential conquerors towards *Pseudomonas aeruginosa*. *Chin. J. Chem.* 2021;39:1093–1103 (b) Zhang GB, Maddili SK, Tangadanchu VKR, et al. Discovery of natural berberine-derived nitroimidazoles as potentially multi-targeting agents against drug-resistant *Escherichia coli*. *Sci. China. Chem.* 2018;61: 557–568.
 - 28 (a) Liu J, Gong YL, Shi JT, et al. Design, synthesis and biological evaluation of novel *N*-[4-(2-fluorophenoxy)pyridin-2-yl]cyclopropanecarboxamide derivatives as potential c-Met kinase inhibitors. *Eur. J. Med. Chem.* 2020;194:112244 (b) Li FF, Zhang PL, Tangadanchu VKR, et al. Novel metronidazole-derived three-component hybrids as promising broad-spectrum agents to combat oppressive bacterial resistance. *Bioorg. Chem.* 2022;122:105718.
 - 29 (a) Lapointe G, Skepper CK, Holder LM, et al. Discovery and optimization of dna gyrase and topoisomerase iv inhibitors with potent activity against fluoroquinolone-resistant Gram-positive bacteria. *J. Med. Chem.* 2021;64:6329–6357 (b) Skepper CK, Armstrong D, Balibar CJ, et al. Topoisomerase inhibitors addressing fluoroquinolone resistance in Gram-negative bacteria. *J. Med. Chem.* 2020;63:7773–7816.
 - 30 (a) Hatzade KM, Ghatole AM, Gaidhane PK, et al. A facile synthesis of some new pyrimidine-2,4,6-triones analogs and their α -D-glucosides P-glycoprotein and antioxidant, antimicrobial study, blood-brain barrier, cytochrome p450 enzyme activity prediction. *Med. Chem. Res.* 2021;30:163–181; (b) Bo YX, Xiang R, Xu Y, et al. Synthesis, biological evaluation and molecular modeling study of 2-amino-3,5-disubstituted-pyrazines as Aurora kinases inhibitors. *Bioorg. Med. Chem.* 2020;28: 115351.